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# Airline ground operations: Schedule recovery optimization approach with constrained resources

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ABSTRACT

Most airlines have established integrated hub and operations control centers for the monitoring and adjustment of tactical operations. However, decisions in such a control center are still elaborated manually on the basis of expert knowledge held by several agents representing the interests of different airline departments and local stakeholders. This article studies a concept which incorporates the situational awareness gained by airport-collaborative decision making into an airline-internal decision support system, such that it integrates all available schedule recovery options during aircraft ground operations. The developed mathematical optimization model is an adaptation of the Resource-Constrained Project Scheduling Problem (RCPSP) and incorporates key features from turnaround target time prediction, passenger connection management, tactical stand allocation and ground service vehicle routing into the airline hub control problem. The model is applied in a case study consisting of 20 turnarounds during a morning peak at Frankfurt airport. Schedule recovery performance (resilience) is analyzed for a set of key performance indicators within multiple scenario instances which contain different resource availability and aim at solving various arrival delay situations. Results highlight that a minimization of tactical cost concurrently reduces average departure delay for flight and passengers while recovery performance is substantially affected when some options are not included in the evaluation process. Thus, our concept provides airlines with an optimization approach for constrained airport resources so that total cost and delay resulting from schedule deviations are reduced, which may benefit strategic schedule planning and improve predictability of operations for local collaborators, such as airport, ground handlers and ATM performance.

## 1. Introduction

Airlines, and in particular large network carriers, typically channel their traffic through large hub airports to profit from economies of scale and scope (Bryan and O'Kelly, 1999). The benefits arise from more potential Origin-and-Destination (O&D) connections which can be offered with fewer aircraft and crews. The downside is a potentially unbalanced usage of airport facilities, given that many flights arrive and later depart at similar times, thereby creating so called "arrival and departure waves". Such densely scheduled operations within a hub bank are necessary to keep total O&D trip times of transfer passenger as short as possible, thus, making these products still sufficiently attractive in comparison to direct O&D flights. However, it leaves airlines with limited room to manoeuvre when they want to build robust schedules, resilient against tactical plan deviations.

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Airline operations, as all forms of mobility, are subject to many stochastic influences (e.g., weather, traffic, technical break-downs, irrational human behavior), which creates the issue of schedule robustness/ resilience. The following approach is *not* envisioned to create schedule robustness but to quantify inherent schedule resilience, given that robustness is mainly understood as a proactive instrument which is pursued at strategic planning level to absorb deviations and create a stable configuration (Wieland and Wallenburg, 2012). Much academic attention has been brought to the subject of allocating schedule buffers in the most cost-efficient way (Stojković et al., 2002; AhmadBeygi et al., 2010; Dück et al., 2012; Aloulou et al., 2013) and to identify weak links in a planned schedule (Rosenberger et al., 2002; AhmadBeygi et al., 2008; Wu and Law, 2019), which translates into a trade-off between planning for profitability and operational robustness (Wu, 2016).

In contrast, resilience is the capacity of a system not only to absorb, but to use existing flexibility and reserves for adapting and recovering from a loss in system performance (Hollnagel, 2013; Francis and Bekera, 2014; Cook et al., 2016; Faber et al., 2017). It complements schedule robustness, given that a schedule made at strategic planning stage (i.e., up to six months prior to the day of operation) cannot account for all potential deviations at tactical level (i.e., on the day of operations), which makes *schedule recovery* as a resilience mechanism inevitable.

### *1.1. Status Quo on Tactical Airline Schedule Recovery*

In 2019, average departure delay in Europe accumulated to 12.8 min per flight, which is still far from the envisioned four-minute ambitions for the year 2035 defined in the "European ATM Masterplan 2020" (European, 2020; Eurocontrol, 2019). According to the EUROCONTROL Performance Review Report, the largest source of primary delay are ground operations (33%); roughly 21% are associated with air traffic flow management (ATFM) regulations; while about 44% of total delay is accounted as reactionary, thus, secondary delay (Eurocontrol, 2019). This distribution highlights the complexity of airline operations during the turnaround (i.e., the ground time between in-block and off-block at an airport stand). However, ATM research had initially been largely focused on in-flight delay reduction, thereby neglecting for long the large share of delay being produced during aircraft ground times. The reason for this is a major difference in paradigms, such that ATFM measures flights as flows of isolated entities, while for an airline, each flight continues on to other flights and is closely connected within the respective (hub) network. Already in 2015, the SESAR ATM Master Plan had identified this ambivalence and recommended more flexibility for airspace users as well as the integration of airports into the ATM network. However, given that none of the established key performance indicators (KPIs) refers to the turnaround (European, 2020), operational improvements are left entirely to the discretion of the airlines themselves.

Airlines have established operations control centers (AOCC) for the monitoring and adjustment of their schedules on the day of operations. Decisions in an AOCC are elaborated manually on the basis of expert knowledge held by several operators which represent the interests of different airline departments and local partners, such as air traffic control, airport and ground handling service providers (Ball et al., 2007). To react to deviations to the original schedule, there have been many research projects aiming for the (partial) automation of AOCC decision-making. Bratu and Barnhart (2006) present one of the first aircraft schedule recovery models which also integrates crew and maintenance constraints and includes passenger cost of delay in the objective function. Kohl et al. (2007) and Clausen et al. (2010) build separate optimisation modules for aircraft, crew and passenger recovery, but conclude that they can hardly be integrated with their full complexity into a real-time decision support system due to evasive computational time. Castro and Oliveira (2011) state their believe that AOCC agents should rather be managers than controllers and propose an agent-based model which develops solutions for highly repetitive tasks and only leaves highly uncertain situations to human operators. Recently, some concepts have emerged which compare cost from additional in-flight fuel consumption for the reduction of potential arrival delays with departure delay cost when connecting flights wait at their airport stands to maintain passenger transfers (Delgado et al., 2016; Marla et al., 2017; Mazzarisi et al., 2019). Montlaur and Delgado (2017) study adaptations of the ground holding problem with special focus on reactionary delay for flights and passengers. They conclude that flight prioritization for optimal arrival sequencing should remain a pre-tactical process (i.e., performed before the day of operations) and might be transferred into the framework of the user-driven prioritization process (UDPP - Pilon et al., 2016) suggested by SESAR, rather than being a stand-alone tactical control option for airlines.

Coming from a macroscopic network perspective, all of these approaches have in common that they reduce ground operations to a single process which has no resilience potential aside from built-in schedule buffers. However, the turnaround is obviously no single process but consists of up to twelve interdependent sub-processes, containing more than 150 individual activities and involving up to 30 different actors (IATA, 2018). Situational awareness among all stakeholders about the current status of operations is envisioned by a milestone concept called Airport-Collaborative Decision Making (A-CDM - Eurocontrol, 2017). However, the benefits from this concept have so far not advanced into a fully operational disruption management concept, given issues with data integrity, consistency and availability. The inherent resilience capacities comprised in ground operations may be minor in comparison to flight cancellations or aircraft re-routings, though, as was found in the Resilience 2050 project, it is especially the reduction of smaller schedule deviations which brings the highest benefit to overall system performance (Eurocontrol, 2016). Those few models which acknowledge the complexity of ground operations, do so for one individual turnaround or for individual resources on a rather microscopic level. Fricke and Schultz (2009) describe the most critical turnaround processes with stochastic time distributions and find that turnaround performance depends on the amount of arrival delay. Kuster et al. (2009) incorporate alternative options for some turnaround sub-processes into their scheduling model to reduce total ground time. Given that the boarding process typically belongs to the critical path of a turnaround, Schultz (2018a,b) develops a stochastic boarding model which can simulate the impact of different procedures on its overall duration. Vidosavljevic and Tosic (2010) integrate flow requirements of multiple airport resources, such as stands, taxiways and ground service equipment (GSE), into a petri net to model interdependencies among multiple turnarounds. Padrón et al.

(2016) build a vehicle routing heuristic for scheduling of all ground-handling vehicles to optimize the time windows dedicated to each turnaround by simultaneously minimizing departure delay. Du et al. (2014) use a similar scheduling model for the efficient routing of towing trucks. Dijk et al. (2019) apply the concept of recoverable robustness to the tactical stand allocation problem which allows aircraft to wait for stands to be vacant or to be re-allocated to other positions, always respecting the consequences of such changes onto passenger transfer times. Ali et al. (2019) find that stand re-allocation can help to minimize missed transfer connections for self-connecting passenger within a low-cost terminal.

### 1.2. Focus and structure

Concluding from the status quo and to the best of our knowledge, there has been no research concept so far which has analysed the inherent resilience potential available to airlines during aircraft ground times. Understanding the full potential of schedule recovery during ground operations is an essential prerequisite for the development of holistic decision support systems. Models on AOCC decision-making focus on large network disruptions and see no resilience capacity during the turnaround aside from schedule buffers, whereas models for ground operations usually analyse isolated problems for a particular turnaround sub-process or airport resource. The approach presented in this article aims at filling this gap in the literature by combining elements from various research problems related to airline schedule recovery and airport logistics, such as:

- Tactical Flight Prioritization;
- Prediction & Control of Turnaround Target Times;
- Aircraft-Specific Delay Metrics;
- Passenger-Oriented Delay Metrics;
- Tactical Crew Control Problem;
- Tactical Stand Allocation; and
- Assignment of Ground Servicing Equipment & Staff.

The focus is to develop a mathematical optimization model which enables the optimal allocation of scarce airport resources to analyse the resilience potential comprised within the turnaround. The model is verified in a case study which contains 20 parallel

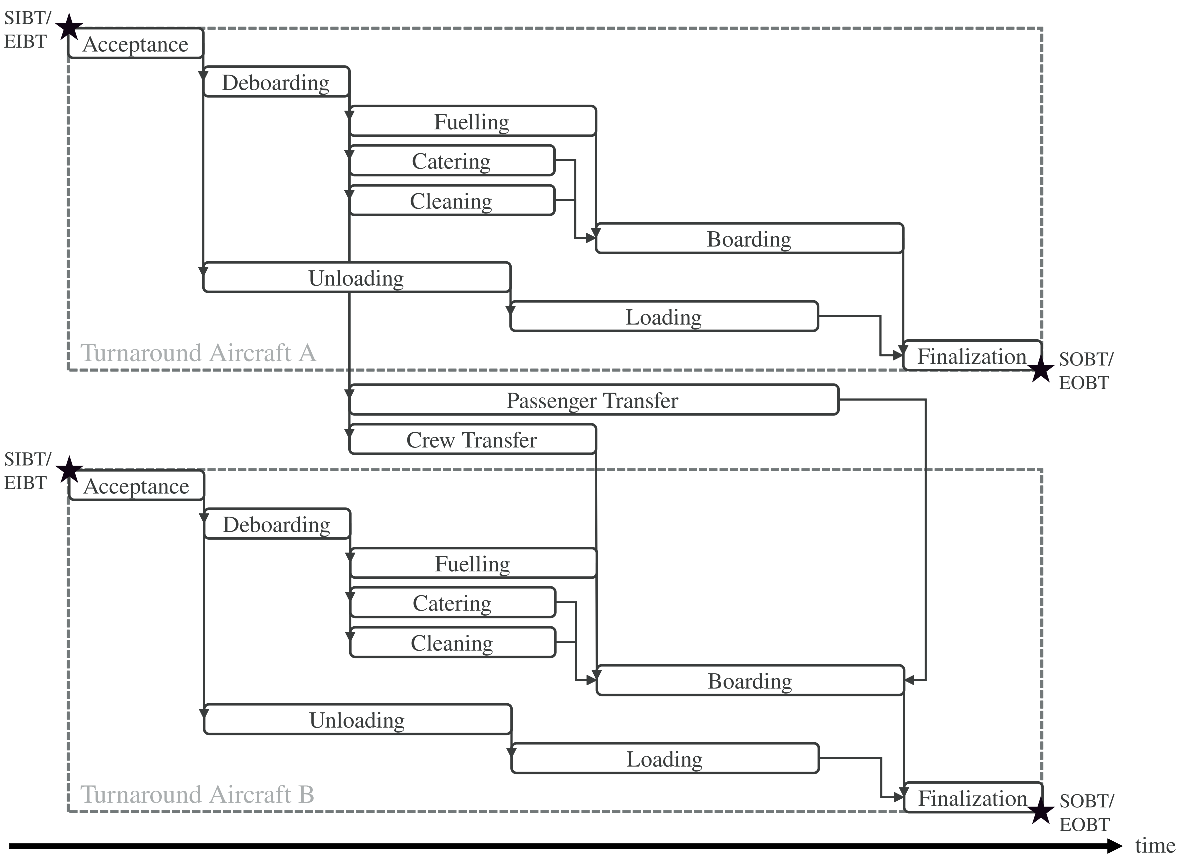



**Fig. 1.** Network graph of turnaround sub-processes.

turnarounds during a morning peak (hub-bank) at Frankfurt airport. Various scenario instances are studied and differ in the availability of airport standard and reserve resources, the level of average arrival delay and the applied airline recovery policy. Section 2 describes the formulation of the optimization model, while Section 3 presents the case study setting as well as the defined scenario instances at Frankfurt airport. Section 4 presents the results from the case study analysis, while Section 5 discusses the results, draws conclusions, and provides an outlook onto future research steps.

## 2. Methodology and Problem Description

The presented method is built on a scheduling model for parallel running turnaround operations. The model itself is an extension of the Resource-Constraint Project Scheduling Problem (RCPSP) such that it considers a set of aircraft $A$ as jobs and airport resources as machines (RCPSP is the general framework of shop scheduling models Nasiri (2013)). Each aircraft $a \in A$ is associated with scheduled and earliest start times of the turnaround $SIBT_a$ and $EIBT_a$ as well as scheduled and earliest finishing times $SOBT_a$ and $eobt_a$. The scheduled values are given from the flight plan, the $EIBT_a$ is adopted from historical flight data and $eobt_a$ is calculated in the scheduling process. The ground handling of all aircraft requires a total set of sub-processes $TP$. Equal turnaround sub-processes are categorized into sub-sets, such that there are sets of aircraft in-block/ acceptance processes $IB$, deboarding processes $DE$, fuelling processes $FU$, catering processes $CA$, cleaning processes $CL$, boarding processes $BO$, unloading processes $UL$, loading processes $LO$ and aircraft finalization processes $FI$. Each single turnaround sub-process $i \in TP$ is characterized by the related aircraft $RA_i$, a variable starting time $s_i$, by links to preceding and succeeding processes determined in the precedence matrix $PM \subseteq TP \times TP$ and by a duration $d_i$. As detailed in Fig. 1, passenger transfer processes $PA$ and crew transfer processes $CW$ build links between two aircraft turnarounds and are assigned to the set of the respective inbound aircraft.

Stochastic process durations have been analysed and case-sensitive probability density functions were determined in prior studies (Fricke and Schultz, 2009; Oreschko et al., 2012; Schultz et al., 2013). From these distributions, mean values are adopted for the respective aircraft type, airport and flight characteristics and used as deterministic parameters in the model. For example, the duration of the cleaning process $d_{CL}$ for an Airbus A320 aircraft is defined by the mean value from a set of functions (1) defined by the trigger parameters aircraft-type $A/C_{type}$, seat-load-factor $SLF_{ARR}$ and distance $Dist_{ARR}$ of the arrival flight (ARR - cf. Fig. 2). Thus, contrarily to common assumptions, the critical path is flexible and can contain different processes for different turnarounds, so that the total turnaround time calculated by the model might differ from official minimum ground times values (Evler et al., 2018).

$$d_{CL} = f(A/C_{type}, SLF_{ARR}, Dist_{ARR}) \tag{1}$$

### 2.1. Objective Function

Airlines establish internal policies which define the objectives of schedule recovery. Three major policy principles can be found in the literature (Rosenberger et al., 2002; Bratu and Barnhart, 2006; Ball et al., 2007; Wu, 2016), which represent divergent interpretations of the typical "wait-or-depart dilemma" in transport nodes:

- Minimize total flight delay time;
- Minimize total passenger delay time; and
- Minimize total cost associated with delay and schedule recovery.

The first principle aims at reducing deviations from schedule on arrival and departure sides, no matter what cost are associated with it. Such a policy implies, e.g., minimal on-stand waiting times for delayed transfer crews, passengers or freight and is likely to result in

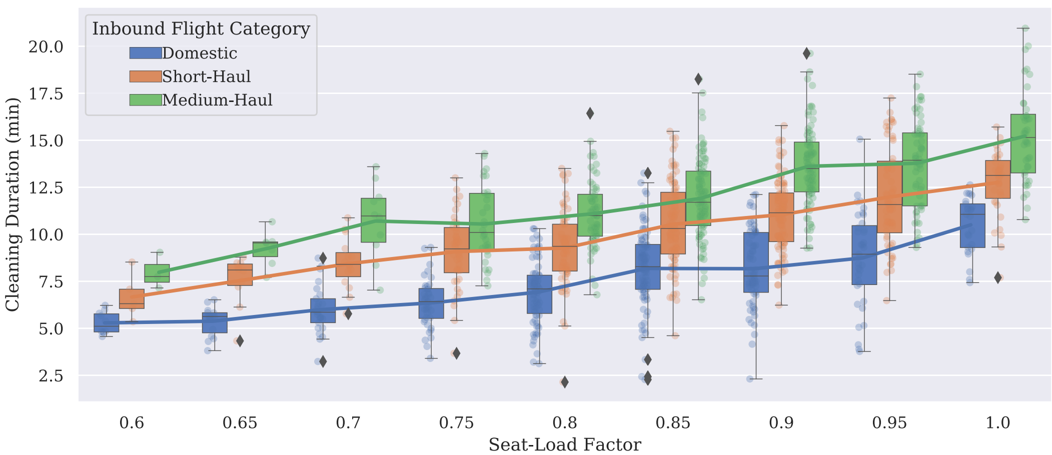



**Fig. 2.** Categorized probability density functions of A320 cleaning duration based on trigger parameters.

**Table 1**
Key Performance Indicators (KPIs) for the analysis of system resilience generated with the developed schedule recovery model.

| KPI | Description | Unit |
|---|---|---|
| total cost of schedule recovery | objective function | Euro |
| flight delay | average departure delay per flight | min |
| passenger delay | average departure delay per passenger | min |

reduced connectivity, meaning that transfer crews need to be replaced by reserves and passengers need to re-booked onto later flights. Although the latter neglects the original customer promise, it is seen as inevitable consequence for maintaining system performance. Montlaur and Delgado (2017) compare the first and the second principle and find that minimizing total flight delays automatically induces reduced passenger delays, given that transfer-passenger profit equally from optimized arrival times, while minimizing passenger delays would increase mean delay values per flight. However, they highlight that these findings should be reassessed with special focus on cost, given that the associated cost are indeed increasing progressively with higher delay due to network propagation effects (Cook, 2015).

Optimizing for minimal total cost can combine the first two objectives and thereby incorporate increasing marginal cost for both, flight and passenger delays. It allows trade-offs between various schedule recovery options for aircraft, crew and passenger itineraries, such that priority can be given to those options which bring the relatively best benefit to the airline (i.e., highest impact at lowest cost). Thus, the third policy principle is adopted as objective for the schedule recovery model. For a comprehensive measurement of system resilience, the prior two principles are anyway included within a set of key performance indicators (KPIs) to analyse trade-offs between cost and delay (see Table 1).

Note that cost associated with standard operations and the original flight plan (which includes opportunity cost from schedule buffers) are omitted given that they cannot be changed in the tactical phase on the day of operations. Further, it needs to be respected that, due to stochastic influences and network effects, not all associated cost from downstream operations can be defined with full accuracy at the moment when a schedule recovery decision needs to be taken. This is considered by adopting progressively increasing reference values from previous studies (see details below) and will be further discussed in Section 5.

### *2.2. Tactical Schedule Recovery Options*

Schedule recovery during ground operations can be categorized into inbound, transit and outbound recovery. Inbound recovery options are applied to reduce arrival delay of an aircraft $a$ (i.e., $EIBT_a > SIBT_a$) in order to maintain critical passenger and crew transfers. Transit recovery options aim at minimizing the ground time of delayed aircraft so that rotational reactionary delay is mitigated. Outbound recovery options influence whether the departure of aircraft $a$ is (further) delayed (i.e., $eobt_a > SOBT_a$) in order to wait for transfer passengers or whether it departs from the airport as scheduled by breaking the respective transfer connections. Note that in some cases an aircraft might already obtain reactionary delay but just not enough to guarantee some passenger transfers. In other cases, additional departure delay might be assigned, considering that, for a holistic optimum, delay could be recovered with buffer capacities on the subsequent flight or by applying dynamic cost indexing (an option which is neglected in this article but discussed in Delgado et al. (2016); Mazzarisi et al. (2019)). Table 2 lists the implemented schedule recovery options with their respective association to inbound, transit and outbound recovery.

Note that our model assumes no tactical cost for stand reallocation, quick passenger transfers and quick de/boarding, given that the model is deemed to calculate with a look-ahead time of at least two hours before the arrival of the first aircraft, which leaves enough time to adapt initial airport resource schedules. Furthermore, airlines typically have long-term service-level agreements with the airport. This means that the airline pays a fixed daily rent to the airport for the sole operation of dedicated stands, so that it can re-

**Table 2**
List of available schedule recovery options for airline ground operations.

| Schedule Recovery Option | Abbreviation | Description | Recovery Category | Tactical Cost |
|---|---|---|---|---|
| Arrival Prioritization | AP | reduction of inbound delay during approach or taxi-in phases | Inbound | n/a |
| Stand Reallocation | SR | adaptation of original stand allocation | Inbound | n/a |
| Quick Passenger Transfer | QP | direct bus transfer for group of transfer passengers | Inbound | n/a |
| Quick-De/Boarding | QB | stand reallocation to remote stand to allow deboarding and boarding via more doors | Inbound/ Transit | n/a |
| Quick-Turnaround | QT | additional cleaning, catering and loading crews + tank protection to allow fuelling while boarding | Transit | Field Data |
| Cancel Passenger Connection | CP | Outbound flight departs without transfer passengers, which are rebooked | Outbound | Cook and Tanner, 2015 |
| Order Standby Crew | SC | Outbound flight departs without transfer crew, which is replaced by stand-by, if available | Outbound | Field Data |
| Departure Delay | DD | outbound flight awaits transfer passengers or crew | Outbound | Cook, 2015 |

arrange its aircraft among these positions. Likewise, the airport guarantees the availability of a certain number of quick passenger transfer busses. Arrival prioritization induces additional workload for an air traffic controller, so that it is rather linked to resource availability than to cost. Tactical cost for quick turnarounds are drawn from field data and include the provision of additional personnel to speed up cleaning, catering and loading processes for selected aircraft and a fee to the fire brigade for tank protection while boarding. Flight delay cost are adopted from reference values per aircraft-type determined by Cook (2015). The cost for cancelling and rebooking passenger transfer connections include basic administrative cost and are supplemented with discounted reference delay cost values per passenger published in Cook and Tanner (2015), such that they consider the additional time until the departure of the next scheduled flight to the passenger's destination (which relates to potential cost for passenger care as well as reimbursement and compensations claims relating to EU regulation 261 (European Union, 2004)) and passenger soft cost. Cost for ordering a standby crew are assumed according to industry wage averages (University of Westminster, 2008) such that two additional hours of paid duty time for the entire crew are calculated because of additional briefing and preparation time. Opportunity cost which may arise because a standby crew is not available for later flights are neglected, given that the model chooses the optimal allocation for the entire hub bank and airlines typically have new standby crews scheduled for each hub bank (i.e., morning, afternoon, evening).

#### 2.2.1. Arrival Prioritization (AP)

For the delay reduction of an arrival flight, the airline can request air traffic control to give priority to the approach or taxi-in process of aircraft $a$. This is determined by $\tau_a = 1$, if the request is granted, 0 otherwise, so that in the prior case the $EIBT_a$ is reduced by a given time $\gamma$. $N^{FP}$ regulates the number of available arrival prioritizations per period. Technically, arrival prioritization is similar to in-flight acceleration or dynamic cost indexing, which would, however, induce additional fuel cost (Delgado et al., 2016) and negative time impacts onto other flights, which are both neglected in the scope of this article.

#### 2.2.2. Stand Reallocation with Quick-Transfers (SR/ QP)

Each aircraft needs to be assigned to either contact stands (CS) at the terminal or remote stands (RS) on the apron, which are all associated with the necessary equipment and personnel for a standard turnaround (typcial RCPSP). Once assigned to a stand, sub-process sequences can be altered and some processes can be accelerated or eliminated depending on the availability of additional airport resources (extended RCPSP, described in further detail in next paragraphs - Kuster et al. (2009); Deblaere et al. (2011)). For each stand there must be a feasible sequence of subsequent assignments starting and ending in idle state defined by a dummy node $S0$. The assignment of aircraft influences the needed transfer times for connecting passengers. Once a transfer process is estimated to become critical, the allocation $\chi_{ak}$ of aircraft $a$ to stand $k$ can be changed. Thus, as detailed in Fig. 3, the needed transfer time $NTT_{kl}$ from stand $k \in RS \cup CS$ to stand $l \in RS \cup CS$ for passengers connecting from aircraft C to B reduces when aircraft C is re-positioned from a remote stand of the set $RS(k = 3)$ to a contact stand of the set $CS(l = 1)$. Note that all other transfer times from and to reallocated aircraft change concurrently, so that the recovery of one transfer process might make others critical.

Additionally, aircraft arriving at a later time need to be considered in the decision process, given that the altered stand occupation might interfere with their initial allocation (marked as red in Fig. 3) and and needs to respect certain transition times $T$ between two subsequent assignments. Transition times are required to prepare equipment and to operate the push back process, if needed. An alternative to stand re-allocation is to assign a quick-transfer bus to a critical passenger transfer process $i$, which is determined by a binary variable $\rho_i = 1$, 0 otherwise. If assigned, the bus transports the associated passenger group via the apron to the respective

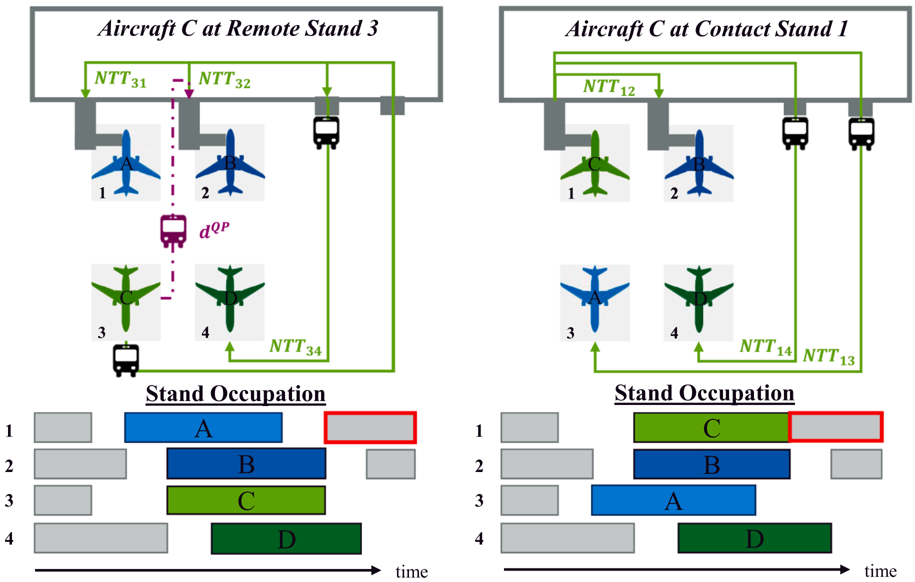



**Fig. 3.** Inbound recovery with tactical stand reallocation and quick passenger transfer.

departure gate so that the needed transfer time is reduced to $d^{QP}$. Such a bus, however, can only accelerate one transfer process at a time and requires the formulation of sequencing constraints starting and ending at a dummy node $P0$. The number of available quick-transfer busses is determined by $N^{QP}$.

#### 2.2.3. Quick-De/Boarding (QB)

Stochastic boarding models have shown that the process can be speeded up significantly when front and rear doors are used simultaneously (Schultz, 2018b). Given that most terminal contact stands only have passenger bridges towards aircraft front doors (one bridge for narrow-body aircraft and two bridges for wide-body aircraft, neglecting A380s), boarding takes often longer at contact stands in comparison to remote stands, where passengers can typically enter the aircraft via stairs at both ends. The same relationship is assumed for the deboarding process and transferred into the model, so that de/boarding durations are reduced by factor $\beta$ when an aircraft $a$ is allocated to a remote stand $k \in RS$. However, longer needed transfer times $NTT_{kl}$ might occur due to the required ride with an apron bus. Note here the interdependency with the stand allocation problem, as the number of parallel quick-de/boarding procedures is limited by the number of available remote stands in $RS$.

#### 2.2.4. Quick-Turnaround (QT)

Typical transit recovery options include additional cleaning, catering and loading crews as well as a tank protection request to flight crew and airport fire brigade whether fuelling can be operated during passenger boarding. The highest impact is achieved when all measures are applied conjointly – which is called Quick-Turnaround – as this was found to guarantee a reduction of the $eobt_a$ even under stochastic influences (Evler et al., 2018). The model assumes a reduction of duration $d_i$ for turnaround sub-processes $i \in CA \cup CL \cup UL \cup LO$ of aircraft $a$ by factor $\alpha$, if $\omega_a = 1$, 0 otherwise. $C_a^{QT}$ represents the associated additional cost for all additional ground handling agents as well as the service fee for tank protection. Similar to quick-transfer, a unit of additional agents can only perform one quick-turnaround at a time and requires a feasible sequence starting and ending at dummy node $T0$. The number of available quick-turnaround units is limited by $N^{QT}$.

#### 2.2.5. Cancellation of Passenger Transfer Connections and Standby Crews (CP/ SC)

The monetary consequences for an airline if a passenger is not arriving at the agreed time at his/her destination (because of reasons the airline is liable for) are prescribed within Regulation EC 261/2004. As compensation and reimbursement need to be initiated by the passenger itself and depend on various side factors, there are certain deductions to be considered when calculating the cost of a missed connection. Thus, as mentioned above, the cost $C_i^{CP}$ for cancelling passenger transfer $i$ are adopted from Cook and Tanner (2015). $\kappa_i = 1$, if a passenger transfer $i$ is cancelled, 0 otherwise. $C_i^{SC}$ are the cost for ordering a standby crew and cancelling a crew transfer $i$. A crew transfer can only be cancelled ($\phi_i = 1$, 0 otherwise), if there is a standby crew available to take over the departure flight. The number of available standby crews is determined by $N^{SC}$.

#### 2.2.6. Departure Delay (DD)

An aircraft $a$ has departure delay if the earliest turnaround finish time $eobt_a$ surpasses the scheduled finish time $SOBT_a$. The resulting total delay is measured with the variable $v_a$. Given the formulation of the model as a mixed-integer linear programme (MILP) and following Stojković et al. (2002), the reference delay cost from Cook (2015) cannot be adopted directly as a curve with increasing marginal delay cost. Instead, the model distributes the total delay $v_a$ over $s \in DS$ linear delay segments. $r_{as}$ is the delay of aircraft $a$ in segment $s$ which has the upper bound $UB_{as}$. As shown in Fig. 4, delay in each segment incurs the respective marginal delay cost $C_{as}^{DD}$ with increasing slopes from segment to segment. The number of segments and respective upper bounds can be adjusted to network-specific airline conditions.

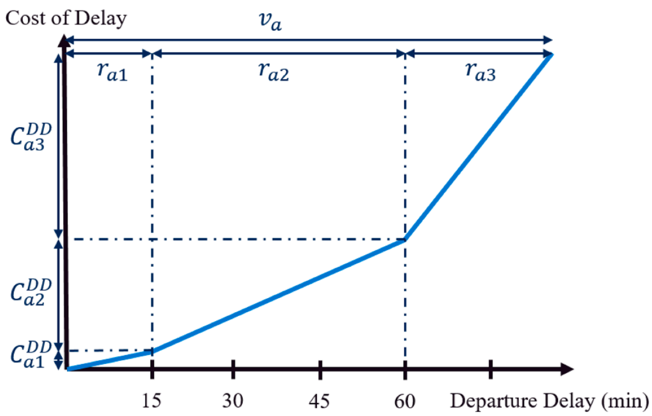



**Fig. 4.** Piecewise linear cost function of departure delay of aircraft a (sketch not-proportional).

### 2.3. Mathematical Model

*Sets:*

| | |
|---|---|
| $A$ | set of all aircraft |
| $TP$ | set of all turnaround processes related to the set of aircraft |
| $IB \subset TP$ | aircraft in-block/ acceptance processes |
| $DE \subset TP$ | deboarding processes |
| $FU \subset TP$ | fuelling processes |
| $CA \subset TP$ | catering processes |
| $CL \subset TP$ | cleaning processes |
| $BO \subset TP$ | boarding processes |
| $UL \subset TP$ | unloading processes |
| $LO \subset TP$ | loading processes |
| $FI \subset TP$ | aircraft finalization processes |
| $PA \subset TP$ | passenger transfer processes |
| $CW \subset TP$ | crew transfer processes |
| $DS$ | set of piecewise-linear delay cost segments |
| $CS$ | set of available contact stands |
| $RS$ | set of available remote stands |
| $P0$ | start and end node for the passenger quick-transfer busses |
| $T0$ | start and end node for the quick-turnaround units |
| $S0$ | start and end node for the sequences of stand allocations |

*Parameters:*

| | |
|---|---|
| $C_a^{QT}$ | cost for assigning a quick-turnaround for aircraft $a$ |
| $C_i^{CP}$ | cost of cancelled passenger transfer $i$ |
| $C_i^{SC}$ | cost of ordering a standby crew instead of crew transfer $i$ |
| $C_{as}^{DD}$ | cost of departure delay of aircraft $a$ for delay level $s$ |
| $N^{QP}$ | number of available quick-transfer busses |
| $N^{QT}$ | number of available quick-turnaround crews |
| $N^{SC}$ | number of available standby cockpit/cabin crews |
| $N^{AP}$ | number of available arrival prioritizations |
| $d_i$ | duration of turnaround process $i$ |
| $NTT_{kl}$ | transfer time between stands $k$ and $l$ |
| $d^{QP}$ | quick-transfer time |
| $\alpha$ | time reduction factor achieved by quick-turnaround |
| $\beta$ | time reduction factor achieved by quick-de/boarding |
| $\gamma$ | time reduction factor achieved by flight prioritization |
| $T$ | transition time between two stand assignments |
| $SIBT_a$ | scheduled in-block time for aircraft $a$ |
| $EIBT_a$ | estimated in-block time for aircraft $a$ |
| $SOBT_a$ | scheduled off-block time for aircraft $a$ |
| $UB_{as}$ | upper bound of delay level $s$ for aircraft $a$ |
| $RA_i$ | related aircraft of turnaround process $i$ |
| $M$ | parameter specific big M |

| | |
|---|---|
| $s_i$ | starting time of process $i$ |
| $v_a$ | total departure delay for aircraft $a$ |
| $r_{as}$ | delay of aircraft $a$ in delay segment $s$ |
| $eobt_a$ | estimated off-block time for aircraft $a$ based on the scheduling process |
| $\omega_a$ | binary variable – equal to 1 if quick-turnaround for aircraft $a$ is assigned, and 0 otherwise |
| $\rho_i$ | binary variable – equal to 1 if quick-transfer for passenger transfer $i$ is assigned, and 0 otherwise |
| $\kappa_i$ | binary variable – equal to 1 if passenger transfer $i$ is cancelled, and 0 otherwise |
| $\phi_i$ | binary variable – equal to 1 if crew transfer $i$ is cancelled, and 0 otherwise |
| $\tau_a$ | binary variable – equal to 1 if flight prioritization for aircraft $a$ is assigned, and 0 otherwise |
| $\chi_{ak}$ | binary variable – equal to 1 if aircraft $a$ is assigned to stand $k$, and 0 otherwise |
| $\psi_{abkl}$ | binary variable – equal to 1 if aircraft $a$ is assigned to stand $k$ and aircraft $b$ to stand $l$,and 0 otherwise |
| $xP_{ij}$ | binary variable for determining the sequence of quick-transfer procedures over all transfer processes $i,j \in PA$ beginning and ending at an dummy node $P0$ |
| $xT_{ab}$ | binary variable for determining the sequence of quick-turnaround procedures over all aircraft $a,b \in A$ beginning and ending at an dummy node $T0$ |
| $xS_{abk}$ | binary variable for determining the sequence of aircraft $a,b \in A$ at stand $k \in \{CS,RS\}$ beginning and ending at an dummy node $S0$ |

*Decision variables:*

$$\min \quad \sum_{a\in A}\left(\sum_{s\in DS} C_{as}^{DD}\ r_{as} + C_a^{QT}\ \omega_a\right) + \sum_{i\in PA} C_i^{CP}\ \kappa_i + \sum_{i\in CW} C_i^{SC}\ \phi_i \tag{2}$$

s.t.

$$s_i \geqslant EIBT_a - \gamma\ \tau_a \qquad \forall i \in IB, a \in RA_i \tag{3}$$

$$s_i + di \leqslant SOBT_a + v_a \qquad \forall i \in FI, a \in RA_i \tag{4}$$

$$v_a = \sum_{s\in DS} r_{as} \qquad \forall a \in A \tag{5}$$

$$r_{as} \leqslant UB_{as} \qquad \forall a \in A; \forall s \in DS \tag{6}$$

$$s_j \geqslant s_i + d_i \qquad \forall i \in IB; \forall j \in TP|(i,j) \in PM \tag{7}$$

$$s_j \geqslant s_i + d_i(1-\chi_{ak}) + \beta\ d_i\ \chi_{ak} \qquad \forall i \in DE \cup BO, a \in RA_i; \forall j \in TP|(i,j) \in PM; \forall k \in RS \tag{8}$$

$$s_j \geqslant s_i + d_i(1-\omega_a) + \alpha\ d_i\ \omega_a \qquad \forall i \in CL \cup CA \cup UL \cup LO, a \in RA_i; \forall j \in TP|(i,j) \in PM \tag{9}$$

$$s_j \geqslant s_i + d_i - M\ \omega_a \qquad \forall i \in FU, a \in RA_i; \forall j \in BO|(i,j) \in PM \tag{10}$$

$$s_j \geqslant s_i + NTT_{kl}\ \psi_{abkl} - M(\rho_i + \kappa_i) \qquad \forall i \in PA, a \in RA_i; \forall j \in BO, b \in RA_j|(i,j) \in PM; \forall k,l \in CS \cup RS \tag{11}$$

$$\chi_{ak} + \chi_{bl} \leqslant 1 + \psi_{abkl} \qquad \forall a,b \in A; \forall k,l \in CS \cup RS \tag{12}$$

$$\chi_{ak} \geqslant \psi_{abkl} \qquad \forall a,b \in A; \forall k,l \in CS \cup RS \tag{13}$$

$$\chi_{bl} \geqslant \psi_{abkl} \qquad \forall a,b \in A; \forall k,l \in CS \cup RS \tag{14}$$

$$\rho_i + \kappa_i \leqslant 1 \qquad \forall i \in PA \tag{15}$$

$$s_j \geqslant s_i + d^{QP} - M(1-\rho_i) \qquad \forall i \in PA, a \in RA_i; \forall j \in BO, b \in RA_j|(i,j) \in PM; \forall k,l \in CS \cup RS \tag{16}$$

$$s_j \geqslant s_i + d_i - M\ \phi_i \qquad \forall i \in CW; \forall j \in BO|(i,j) \in PM \tag{17}$$

$$\sum_{a\in A\cup S0} xS_{abk} = \chi_{bk} \qquad \forall b \in A; \forall k \in CS \cup RS \tag{18}$$

$$\sum_{b\in A\cup S0} xS_{abk} = \chi_{ak} \qquad \forall a \in A; \forall k \in CS \cup RS \tag{19}$$

$$s_i \geqslant eobt_a + T - M(1 - xS_{abk}) \qquad \forall a \in A; \forall b \in A \cup S0; \forall i \in IB, b \in RA_i; \forall k \in CS \cup RS \tag{20}$$

$$\sum_{k \in CS \cup RS} \chi_{ak} = 1 \qquad \forall a \in A \tag{21}$$

$$\sum_{i \in PA \cup P0} xP_{ij} = \rho_j \qquad \forall j \in PA \tag{22}$$

$$\sum_{j \in PA \cup P0} xP_{ij} = \rho_i \qquad \forall i \in PA \tag{23}$$

$$s_j \geqslant s_i + d^{QP} + T - M(1 - xP_{ij}) \qquad \forall i \in PA, a \in RA_i; \forall j \in PA \cup P0, b \in RA_j; \forall k, l \in \{CS, RS\} \tag{24}$$

$$\sum_{i \in PA \cup P0} xP_{P0 \to i} \leqslant N^{QP} \tag{25}$$

$$\sum_{a \in A \cup T0} xT_{ab} = \omega_b \qquad \forall b \in A \tag{26}$$

$$\sum_{b \in A \cup T0} xT_{ab} = \omega_a \qquad \forall a \in A \tag{27}$$

$$s_i \geqslant eobt_a + T - M(1 - xT_{ab}) \qquad \forall a \in A; \forall b \in A \cup T0; \forall i \in IB, b \in RA_i \tag{28}$$

$$\sum_{a \in A \cup T0} xT_{T0 \to a} \leqslant N^{QT} \tag{29}$$

$$\sum_{a \in A} \tau_a \leqslant N^{AP} \tag{30}$$

$$\sum_{i \in CW} \phi_i \leqslant N^{SC} \tag{31}$$

The objective function (2) minimizes the total tactical cost incurred by departure delays and quick-turnarounds across all aircraft as well as for cancelled passenger and crew transfers. Constraints (3) ensure that the start of a turnaround for aircraft $a$ is not scheduled before its estimated arrival time $EIBT_a$, which can be pulled forward by an arrival prioritization. Constraints (4) determine the total delay $v_a$ of aircraft $a$, which is distributed over the respective delay segments (5), whereby the delay in each segment is limited by (6). Scheduling constraints (7)–(11) ensure the precedence relationships among turnaround sub-processes. Constraints (7) represent the standard form and are valid for processes after aircraft in-block/ acceptance. Constraints (8) schedule processes starting after de-/

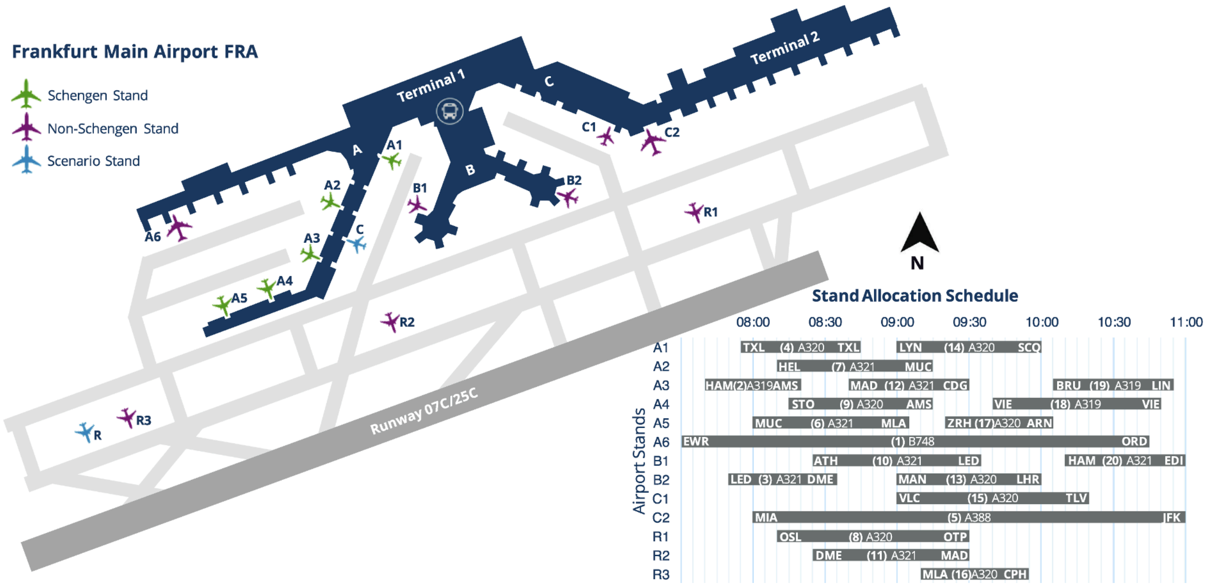



**Fig. 5.** Map of Frankfurt airport with aircraft positions for case study.

**Table 3**
Simulated passenger transfer matrix between aircraft.

| Transfers | to | Aircraft | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| from Aircraft | Origin | Destination SIBT/SOBT | ORD 10:45 | AMS 08:20 | DME 08:35 | TXL 08:45 | JFK 11:00 | MLA 09:05 | MUC 09:15 | OTP 09:30 | AMS 09:15 | LED 09:35 | MAD 09:30 | CDG 09:30 | LHR 10:00 | SCQ 10:00 | TLV 10:20 | CPH 09:55 | STO 10:05 | VIE 10:50 | LIN 10:55 | EDI 11:00 |
| 1 | EWR | 07:30 | - | 48 | 25 | 21 | - | 4 | 4 | 4 | - | 4 | - | - | - | - | 5 | - | 12 | 15 | 2 | - |
| 2 | HAM | 07:40 | 3 | - | 20 | - | 4 | 2 | 6 | 5 | - | 8 | 4 | 3 | - | 1 | 3 | - | - | - | 3 | 2 |
| 3 | LED | 07:50 | - | - | - | - | 39 | 30 | - | - | - | - | 8 | 14 | - | 7 | - | - | - | - | 5 | - |
| 4 | TXL | 07:55 | 11 | - | - | - | 16 | 13 | - | - | 16 | - | 10 | 5 | 9 | 3 | - | - | - | - | 2 | 6 |
| 5 | MIA | 08:00 | - | - | - | 44 | - | 21 | 7 | 35 | - | 33 | - | - | - | - | 7 | 38 | 35 | 8 | 10 | - |
| 6 | MUC | 08:00 | 5 | - | - | 5 | 1 | - | - | - | 23 | - | 5 | 16 | 3 | 4 | - | 14 | 4 | - | - | 4 |
| 7 | HEL | 08:10 | 15 | - | - | - | 9 | 6 | 14 | - | - | - | 7 | 16 | 10 | 9 | - | - | - | - | 6 | - |
| 8 | OSL | 08:10 | - | - | - | - | - | - | 15 | 20 | - | - | 9 | 17 | - | 1 | 15 | - | - | 6 | - | - |
| 9 | STO | 08:15 | - | - | - | - | 8 | 3 | 24 | - | - | - | 8 | 3 | 16 | 13 | 1 | - | - | - | 10 | - |
| 10 | ATH | 08:25 | 15 | - | - | - | 3 | - | - | - | 4 | - | 13 | 12 | 10 | 13 | - | 5 | 1 | - | - | 3 |
| 11 | DME | 08:25 | 2 | - | - | - | 27 | - | - | - | 24 | - | 15 | 7 | 4 | 8 | - | - | - | - | 6 | 12 |
| 12 | MAD | 08:40 | - | - | - | - | - | - | - | 10 | - | 24 | - | - | - | - | 10 | 18 | 3 | 15 | - | - |
| 13 | MAN | 09:00 | - | - | - | - | - | - | - | - | - | - | - | - | - | 9 | 44 | - | - | 28 | 9 | - |
| 14 | LYN | 09:00 | 24 | - | - | - | - | - | - | - | - | - | - | - | - | - | - | 8 | 27 | 12 | - | 11 |
| 15 | VLC | 09:00 | - | - | - | - | - | - | - | - | - | - | - | - | - | - | 19 | 16 | 10 | 10 | - | 13 |
| 16 | MLA | 09:10 | 34 | - | - | - | 33 | - | - | - | - | - | - | - | 5 | - | - | 2 | 2 | - | - | 10 |
| 17 | ZRH | 09:20 | 12 | - | - | - | 9 | - | - | - | - | - | - | - | - | - | - | - | 18 | - | - | 28 |
| 18 | VIE | 09:40 | 33 | - | - | - | 29 | - | - | - | - | - | - | - | - | - | - | - | - | - | - | 6 |
| 19 | BRU | 10:05 | - | - | - | - | 7 | - | - | - | - | - | - | - | - | - | - | - | - | 17 | 18 | - |
| 20 | HAM | 10:10 | - | - | - | - | 17 | - | - | - | - | - | - | - | - | - | - | - | - | 6 | 28 | 3 |

boarding, whereby de-/boarding durations are reduced if aircraft $a$ is positioned on a remote stand $k \in RS$. Constraints (9) reduce the duration of catering, cleaning, and cargo processes if a quick-turnaround unit is assigned. In case of quick-turnaround, constraints (10) further allow the parallel execution of fuelling and boarding (i.e., with tank protection), which otherwise need to be scheduled sequentially. Constraints (11) determine the needed transfer time $NTT_{kl}$ for passengers connecting from aircraft $a$ on stand $k$ to aircraft $b$ and stand $l$. Note that the last term restricts the time constraints to cases when there is no quick-transfer or cancelled connection. Further note that the original quadratic formulation is linearized in (12)–(14) for the solution with most standard solvers. Constraints (15) ensure that there is either a quick-transfer or a cancellation. Constraints (16) induce a reduced transfer duration $d^{QP}$ once a quick-transfer is assigned to a passenger transfer process. Constraints (17) schedule crew transfers, which are neglected in case a crew can be replaced by a standby. MTZ sequencing constraints (18)–(20) consider that all stands can only be occupied by one aircraft at a time, thereby respecting transition times $T$ between two subsequent assignments (20). Constraints (21) ensure that all aircraft are allocated to only one stand respectively. MTZ sequencing constraints (22)–(24) ensure that a quick-transfer bus can only be assigned to one passenger transfer process at a time, respecting that (25) determines the number of available buses $N^{QT}$. Equally, MTZ constraints (26)–(28) regulate the number of parallel quick-turnarounds, whereby (29) defines the maximum number of available units $N^{QT}$. Finally, constraints (30) and (31) determine the number of possible arrival prioritizations $N^{AP}$ and available stand-by crews $N^{SC}$ for the considered period.

## 3. Implementation

The model from Section 2.3 is implemented in a case study which consists of 20 turnarounds during a morning hub-bank (7:30 to 11:00 a.m. local time) at Frankfurt Airport (FRA). The respective flight plan data are extracted from the summer schedule 2017 of a local hub carrier. Within the case study setting, different scenario instances with changing resource availability are run for ten arrival delay situations (days) during the summer 2017. Thereby, for each situation, the respective actual in-block times (AIBTs) are adopted as estimate values $EIBT_a$ for all 20 aircraft (given a setting with two hour look-ahead time until arrival) to study how the model copes with different levels of average arrival delay.

### 3.1. Airport Setting

Initial stand allocation and scheduled block times (SIBT and SOBT) of all 20 aircraft are adopted from one day of operations during the summer schedule 2017 and have been cross-validated with operations on other days. While SIBT and SOBT values (including schedule buffers) are fairly stable, stand allocation varies on a daily basis, such that only few aircraft are positioned at the same gate for the entire season. Thus, a day was selected which comprised a stand allocation spanning evenly across all airport areas normally used by the carrier. This includes five contact stands in Terminal 1A (stands A1–A5 in Fig. 5), which are reserved for flights from and to Schengen countries only. Contact stands with special security and customs areas (stands A6, B1-B2, C1-C2) can also operate flights from and to Non-Schengen countries, stands A6 and C2 being predominantly used for intercontinental flights with wide-body aircraft. Stands R1 to R3 are remote stands where passengers need to be transferred with apron busses via the central bus station (marked with the bus icon in Fig. 5). Remote stands can be used for flights from and to Schengen and Non-Schengen countries. Schengen contact stand C and remote stand R were not occupied on the selected day and are introduced for the scenarios explained in subsection 3.3. Needed transfer times between stands $NTT_{kl}$ are determined according to actual walking distances inside the terminal and additional driving distances on the apron for remote positions (considering a surplus for bus loading and unloading). In contrast, quick passenger transfer times $d^{QP}$ consider only driving distances on the apron, given that passengers are directly fetched at the arrival gate and transported to their departure gate.

Fig. 5 further shows the initial stand allocation of all 20 scenario aircraft with their respective arrival and departure destinations in three-letter IATA codes, aircraft type and aircraft number within the scenario. The case study comprises two wide-body aircraft operating long-haul flights and 18 narrow-body aircraft operating across Europe.

**Table 4**
Description of available schedule recovery resources per scenario instance.

| ID | Scenario Description | Inbound Recovery Resources | | | Transit | Outbound | | |
|---|---|---|---|---|---|---|---|---|
| | | AP | SR/ QB | QP | QT | CP | SC | DD |
| BL | baseline (no recovery, only buffers) | - | - | - | - | - | - | avail. |
| IB | only inbound recovery | 1 prio. | 10 CS/ 3 RS | 1 bus | - | - | - | avail. |
| TR | only transit recovery | - | - | - | 1 unit | - | - | avail. |
| OB | only outbound recovery | - | - | - | - | avail. | 1 crew | avail. |
| CB | combined inbound, transit and outbound recovery | 1 prio. | 10 CS/ 3 RS | 1 bus | 1 unit | avail. | 1 crew | avail. |
| CB + C | combined recovery with extra inbound resources (contact stand C) | **2** prio. | **11** CS/ 3 RS | **2** busses | 1 unit | avail. | 1 crew | avail. |
| CB + R | combined recovery with extra inbound resources (contact stand R) | **2** prio. | 10 CS/ **4** RS | **2** busses | 1 unit | avail. | 1 crew | avail. |
| CB + QT | extra transit steering (extra quick turnaround crew) | 1 prio. | 10 CS/ 3 RS | 1 bus | **2** units | avail. | 1 crew | avail. |

### 3.2. Passenger and Crew Transfer Connections

Passenger and crew connections are simulated once and kept equal for all scenario instances, given the lack of actual itineraries. This ensures comparability between instances when resource availability and arrival delay are variable. The simulation is thereby constructed to resemble potential itineraries which adhere to the average connection index (55% transfer passengers) and minimum connecting time in FRA (45 min), typical airline seat-load-factors (85%) and avoid extreme detours (e.g., a passenger from Malta is unlikely to connect via FRA to Milan). Connections do not differentiate between different booking classes or passenger loyalty states. As a result, 145 passenger transfers are generated between all 20 aircraft (see flow dimensions in Table 3). Crew swaps are scheduled between aircraft 10 and 11 (crew 10: DME-LED; crew 11: ATH-MAD) and aircraft 14 and 15 (crew 14: LYN-TLV; crew 15: VLC-SCQ), while crews from long-haul destinations EWR and MIA terminate in FRA. All other 14 crews are scheduled to continue their flight duties with their respective inbound aircraft.

### 3.3. Resource-Constrained Scenarios

Airport and ground handling resources are the enablers for schedule recovery options during aircraft ground times. Therefore, a set of scenarios studies the impact of changing resource availability on the overall recovery performance. Different scenario instances are designed, such that there is a baseline instance with no recovery options available *BL* and further instances where only inbound (*IB*), transit (*TR*) and outbound (*OB*) recovery options are available. The combination of these three recovery categories is described as instance *CB*. Additional combined recovery instances $CB + C$, $CB + R$ and $CB + QT$ contain a higher amount of inbound and transit recovery resources respectively, in order to quantify how much additional resilience can be achieved in comparison to *CB*. An overview of available resources for each scenario instance is listed in Table 4 and corresponds to the sets and parameters $N^{AP}$ for arrival prioritizations, *CS* for contact stands, *RS* for remote stands, $N^{QP}$ for quick passenger transfer busses, $N^{QT}$ for quick turnaround crews and $N^{SC}$ for standby crews.

Outbound recovery is less resource-oriented than inbound and transit recovery. It depends on the implemented recovery policy of the airline, which focuses on cost evaluations for departure delay and missed passenger transfer connections. In our case, the baseline recovery policy is to maintain all passenger transfer connections (which is the reason why the *DD* option is available across all scenario instances in Table 2), even if this may induce high departure delays onto other flights. The respective marginal delay cost per minute are interpolated from reference values per aircraft type defined by Cook (2015), such that they include passenger soft-cost, costs related to additional crew wages and required maintenance as well as reactionary delay cost. Thus, marginal cost remain stable per delay segment, whereby segment $s = 1$ contains the first 15 min of delay ($UB_{a1} = 15$) with the lowest slope, given that such deviations are not officially registered as "delay" and can be compensated more easily (e.g., by implemented schedule buffers). The second segment $s = 2$ includes delay values between 16 and 60 min ($UB_{a2} = 45$) with an higher slope to resemble increasing reactionary cost from potential downstream deviations. Departure delays in segment $s = 3$ (i.e., above 60 min) are measured with the highest slope, given that such delays are almost certainly causing a large number of downstream schedule disruptions.

As described above, cost for cancelled passenger transfers are adopted from discounted reference values defined by Cook and Tanner (2015), such that they consider the additional trip time with the next flight towards the respective destination and the corresponding compensation with respect to EU regulation 261. The discounted values are multiplied by the number of passengers per transfer connection (cf. Table 3). Cost per quick-turnaround are based on data from the field (i.e., service level agreements) and set to be 500 Euro per assignment. A quick-turnaround reduces the respective sub-processes by 30% (i.e., $\alpha = 0.7$; Evler et al. (2018)), independent of the aircraft-type. Quick-de/boarding applies the same reduction factor (i.e., $\beta = 0.7$; Schultz, 2018a). The time benefit of arrival prioritisation is determined at $\gamma = 5min$ and neglects potential de-prioritisation impacts on other aircraft, which is discussed in Section 5. As mentioned in subsection 2.2.5, cost for ordering a standby crew include additional wage cost equivalent to two extra hours of duty time, which equals roughly 1000 Euro for an A320 aircraft with two flight crew and four cabin crew members (University of Westminster, 2008).

The set of scenarios is tested on ten arrival delay situations as observed during ten different operational days in the summer season 2017 of a local hub carrier in FRA. All ten situations were selected to represent constellations with small to medium deviations,

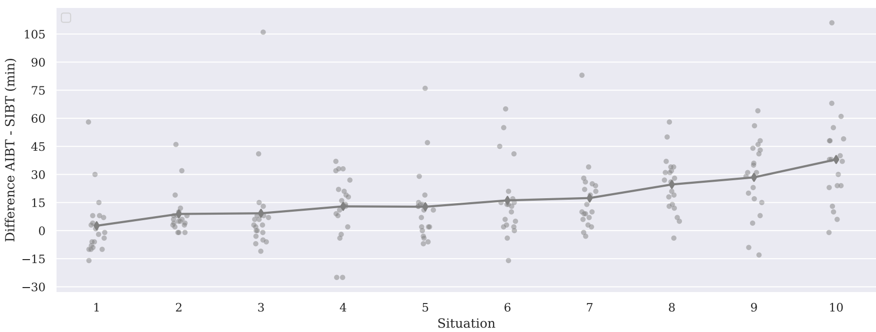



**Fig. 6.** Dispersion of arrival times for 20 aircraft according to ten actual situations during the summer season 2017.

**Table 5**
Computing times to determine optimal solution (percentage values display remaining gap after 3600s).

| - | BL | IB | TR | OB | CB | CB + C | CB + R | CB + QT |
|---|---|---|---|---|---|---|---|---|
| Day 1 | 5.73 s | 3,15% | 5.82 s | 4.93 s | 1436.04 s | 1868.39 s | 2151.77 s | 689.57 s |
| Day 2 | 5.11 s | 11,20% | 4.97 s | 4.50 s | 1156.63 s | 0,96% | 0,85% | 977.02 s |
| Day 3 | 8.91 s | 25,51% | 7.00 s | 5.14 s | 2503.21 s | 2427.91 s | 1560.50 s | 2559.57 s |
| Day 4 | 3.14 s | 1,60% | 4.51 s | 4.39 s | 12,28% | 3,58% | 2644.99 s | 6,35% |
| Day 5 | 5.68 s | 0,21% | 4.83 s | 4.85 s | 1839.55 s | 2844.20 s | 1524.47 s | 1055.91 s |
| Day 6 | 5.61 s | 2,66% | 4.81 s | 4.72 s | 2457.81 s | 2595.58 s | 2943.90 s | 1484.94 s |
| Day 7 | 5.70 s | 3,24% | 6.07 s | 4.95 s | 2,61% | 2820.56 s | 2859.69 s | 3,67% |
| Day 8 | 4.87 s | 6,02% | 14.96 s | 4.50 s | 11,37% | 0,38% | 3,39% | 10,55% |
| Day 9 | 5.12 s | 6,39% | 4.47 s | 4.41 s | 10,33% | 3,95% | 5,87% | 11,10% |
| Day 10 | 6.19 s | 6,73% | 5.62 s | 5.49 s | 2661.71 s | 1884.64 s | 3357.18 s | 2421.32 s |

including increasing average arrival times between 3 and 38 min higher than scheduled ones. Average delay was chosen to be the driving element for the selection because of its nomination as key performance indicator (KPI) in the SESAR performance evaluation framework (European, 2020). Note that pairs of situations (situations 2 & 3, 4 & 5, 6 & 7, 8 & 9) were selected which contain similar average values but very heterogeneous dispersion (cf. Fig. 6). Heavy delay situations were excluded from the analysis, given that they came along with the cancellation of at least one flight.

## 4. Results

The mathematical model was implemented in a Java environment and solved with Gurobi package 8.0 on a 24-kernel CPU with 32 GB RAM. The solution process was aborted after one hour in case that optimality of a solution could not be proven and there was still a gap between upper and lower bounds. Table 5 details the required run times for all scenario instances and the respective gaps in case optimality could not be proven after 3600 s.

The results from all scenario instances defined in subsection 3.3 are bench-marked against the baseline instance (*BL*) where no schedule recovery options are available such that all scheduled process interdependencies are maintained. Thus, in *BL*, arrival delay can only be absorbed if there are strategic schedule buffers comprised within scheduled ground times of an aircraft and along scheduled passenger and crew transfer connections.

### *4.1. Objective Function (Total Cost)*

Fig. 7 shows absolute total cost per arrival delay situation and scenario as well as relative cost benefits achieved by the application of the respective recovery options. It is most obvious that aircraft with more than 100 min of arrival delay seem to be major cost drivers if all transfer connections were maintained (*BL*). Because of such outlier values, variance of arrival delay seems to have a much higher impact on airline cost than average arrival delay, as can be observed in situations 2 and 3 (both with 9 min average arrival delay), 4 and 5 (both 13 min) as well as 6 and 7 (16 and 17 min respectively).

Judging from the results of separate inbound (*IB*), transit (*TR*) and outbound (*OB*) recovery, the latter scenario generates on average the lowest cost for the selected ten situations, although it also has the most volatile performance. This volatility can be explained by the fact that different transfer connections are impacted in each scenario, such that the number of affected passengers and the associated rebooking and compensation cost are very heterogeneous and depend on the respective network constellation. Thus, in some situations, inbound recovery is more efficient, while transit recovery generally has a very limited impact on total cost. Again, relative cost reductions do not correlate with the amount of average arrival delay, given that delay almost triples from situation 5 to 10 without reducing relative cost impacts of inbound and transit recovery.

Combining the recovery options of all three categories (*CB*) compensates the individual weaknesses and provides generally lower cost. For each situation, at least 50% of total cost can be saved in comparison to *BL*, whereas, across all ten situations, total cost can be reduced by 77.6%. Additional airport and ground handling resources in instances $CB + C$, $CB + R$ and $CB + QT$ are able to generate an additional benefit of 6.3%, 7.5% and 7.3% respectively in comparison to *CB*, and 79.0%, 79.3% and 79.2% respectively in comparison to *BL*. Note that these benefits are mainly obtained in situations with already low total cost, whereas in situations 3 and 10 they have only very limited impact (cf. Fig. 7). Considering that all three instances use the additional flexibility provided by extra resources to generate similar cost reductions, strategic cost-benefit-assessments should be made. These should evaluate the strategic (opportunity) cost for each additional resource (i.e., potentially higher fees in the service level agreement with the ground handler/ airport) with the reduced tactical cost per situation for the entire season. Although tactical cost are reduced by on average 1500 to 1800 Euro in the studied situations, other days may obtain less deviations, such that additional resources (which are paid for in long-term contracts) are not required.

### *4.2. Flight and Passenger Delay*

As mentioned in Section 2.1, some schedule recovery models in the literature have optimized for departure delay per flight or passenger (Montlaur and Delgado, 2017). Thus, both objectives have been adopted as KPIs and it is studied how a minimal total cost

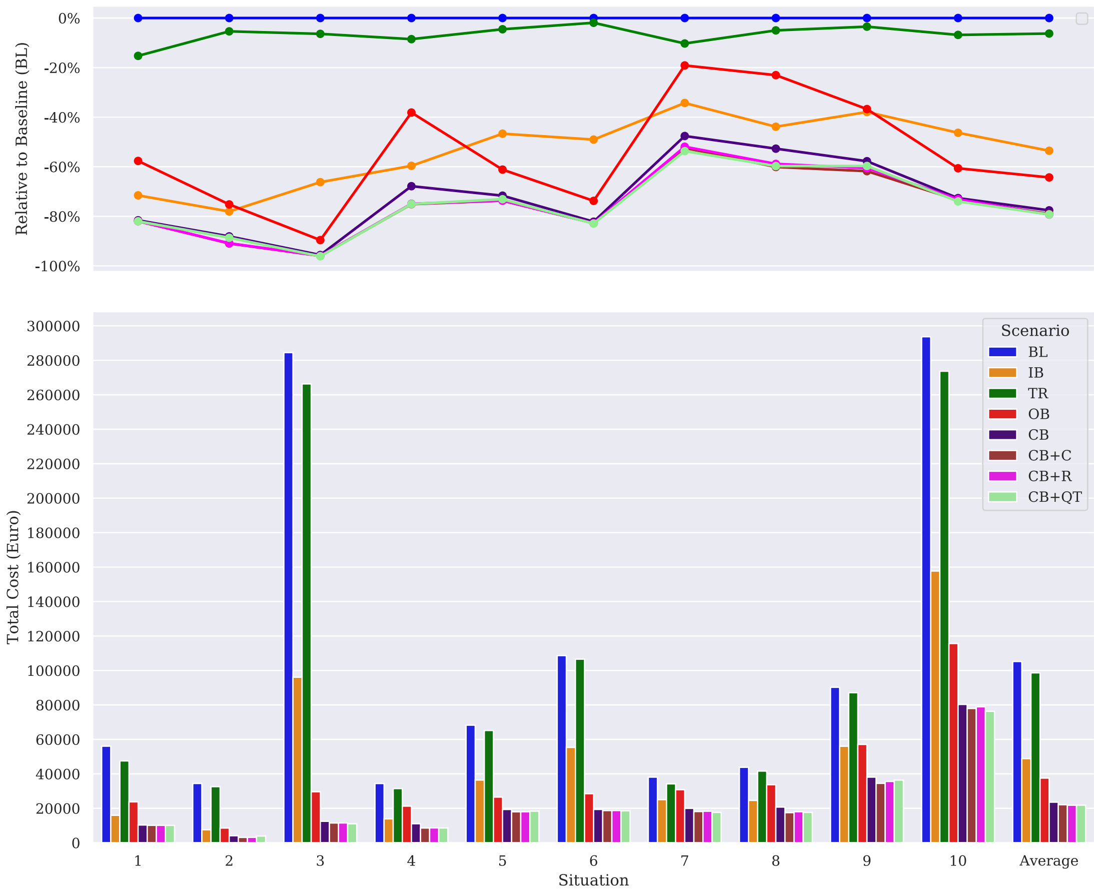



**Fig. 7.** Results of schedule recovery model (objective function). Comparison of total cost per scenario for different arrival delay situations.

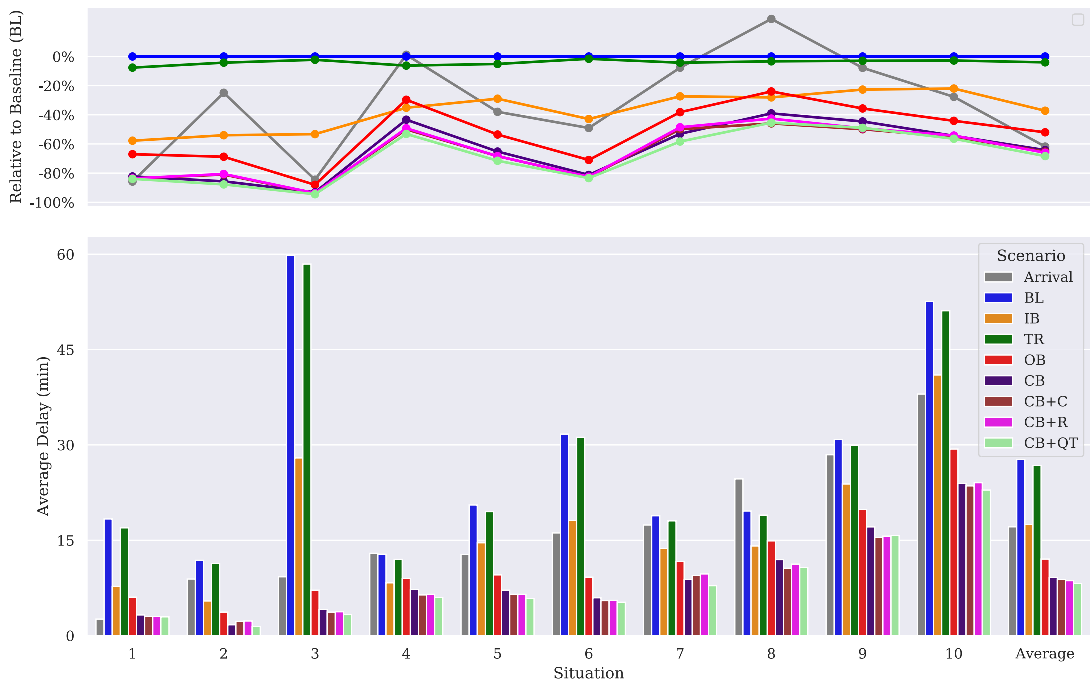



**Fig. 8.** Calculated average departure delay of all flights as part of the objective function.

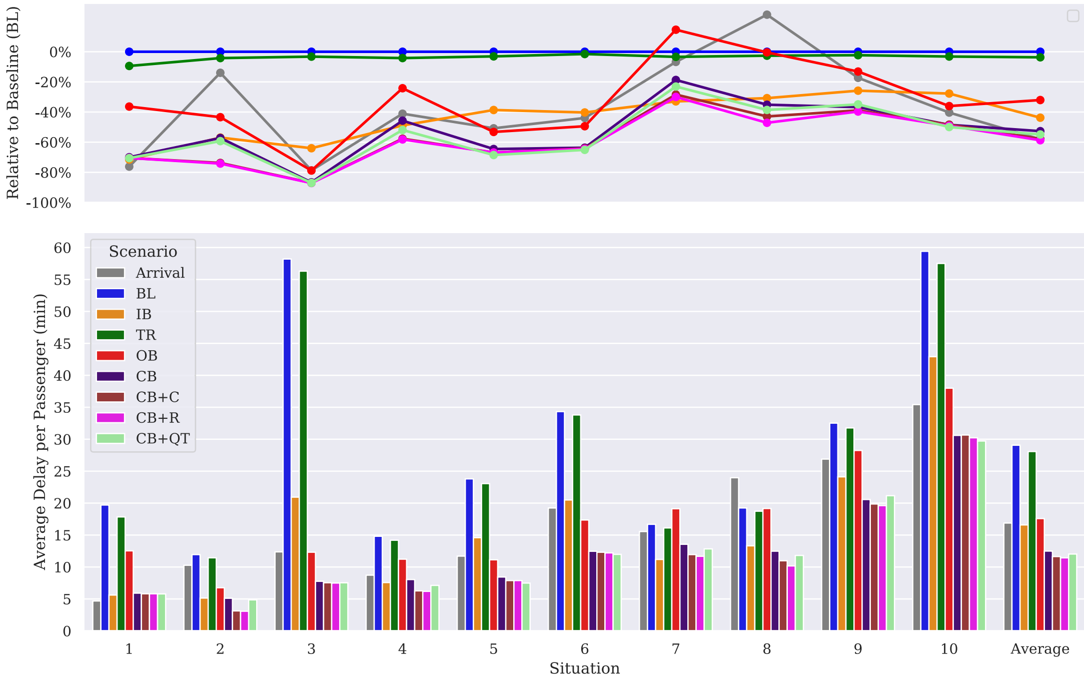



**Fig. 9.** Calculated average departure delay per passenger as part of the objective function.

objective function influences these two values when they are both translated into cost and incorporated into the objective function. Given that airlines typically consider service quality within their decision-making process, results for the two delay KPIs are further related to common airline-internal performance indicators, such as *on-time performance* (OTP), which measures the share of flights which departed with a maximum deviation of 15 min (which is officially regarded as "no delay"), and *transfer ratio*, which is the share of successfully connecting passengers.

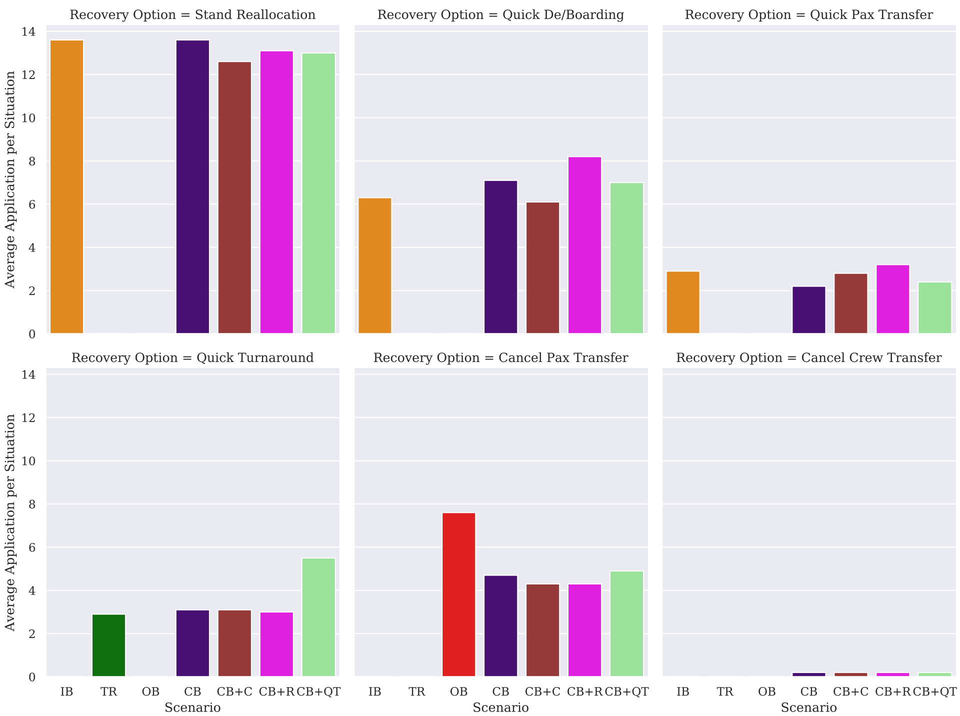


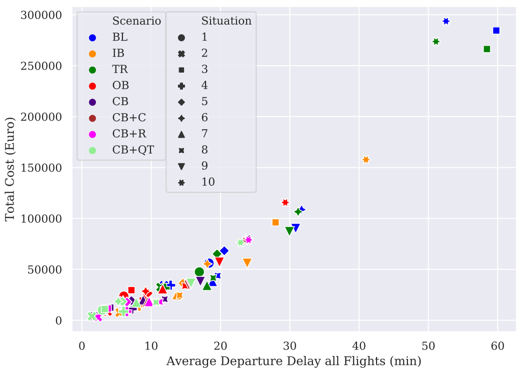

**Fig. 10.** Average number of recovery options applied per day in all scenario instances.

Fig. 8 reveals a similar data pattern for average departure delay as seen in total cost (cf. Fig. 7). Hence, there is a strong correlation along a progressive regression curve (cf. Fig. 11a), such that, with no recovery (*BL*), high arrival delay outliers result in high average departure delay, due to non-rotational reactionary delays (cf. situations 3 and 10). With the exception of situations 4 and 8, where the average arrival delay is maintained or even reduced on the departure side, delay significantly increases for the eight remaining situations from an average of 17.1 min to 27.7 min during ground operations if not recovered (*BL*). Thus, only 36.5% of all flights across all ten situations would depart ”on-time” (with less than 15 min of departure delay).

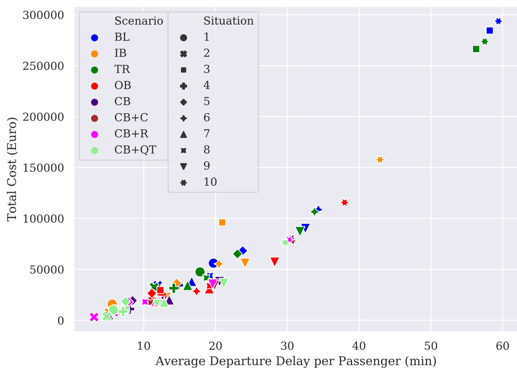
Considering separate recovery categories, outbound recovery (OB) naturally generates the lowest departure delay values (roughly 50% reduction on average) due to increased elimination of transfer links between turnarounds (cf. Fig. 10). This helps to isolate deviations so that arrival delay levels are maintained on the departure side without inducing much additional reactionary delay. Alone with this option, on-time performance (OTP) could be doubled to 70%.

In the combined recovery scenario (*CB*), average departure delay can be reduced by 64.2% to an average of 9.1 min. Thereby, OTP increases to 77%, despite on average less eliminated passenger transfers per situation (cf. Fig. 10). Scenario $CB + C$, which incorporates more inbound recovery resources and an additional contact Schengen stand ($C$), generates an average delay reduction of 65.2%, whereas in scenario $CB + R$ an additional remote stand ($R$) reduces average delay by 66%. The latter can be explained by the possibility to operate more quick-de/boarding procedures (cf. Fig. 10), which reduces the total turnaround time (cf. Section 2.2.3) but also increases transfer times, such that more quick passenger transfers need to be operated (cf. Fig. 10). With 80% OTP, highest resilience for the aircraft schedule is achieved in scenario $CB + QT$, where the number of operated quick-turnarounds increases by 40% due to the availability of a second crew (cf. Fig. 10), which reduces average departure delay by 68.3% to 8.2 min.

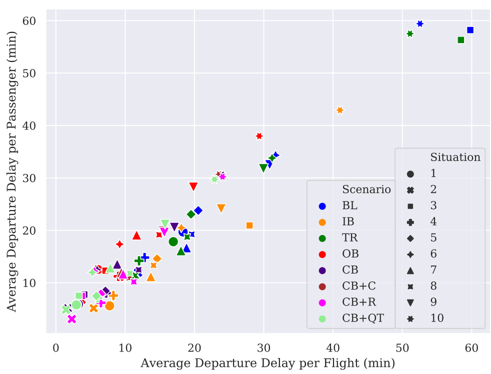
In terms of average departure delay per passenger, the model generates a benefit of 52.7% in *CB* (cf. Fig. 9). Thus, while passenger delay exceeded flight delay by just 1.0 min in *BL*, this difference increases to 3.4 min in *CB* (12.5 min vs. 9.1 min - cf. Fig. 8). This effect is also visible in Fig. 11b, where the regression curve is slightly elongated along the x-axis in comparison to Fig. 11a, whereas in Fig. 11c, the delay reduction in all *CB* scenarios is greater along the x-axis (flight delay) than along the y-axis (passenger delay). This confirms findings from previous studies, such that an integrated assessment of recovery options can reduce both values but benefits for flight delay exceed those for passenger delay, given that re-booked passengers have extended waiting times until the next flight on the same route (Delgado et al., 2016; Montlaur and Delgado, 2017). Note, however, that departure delay for non-connecting passengers in



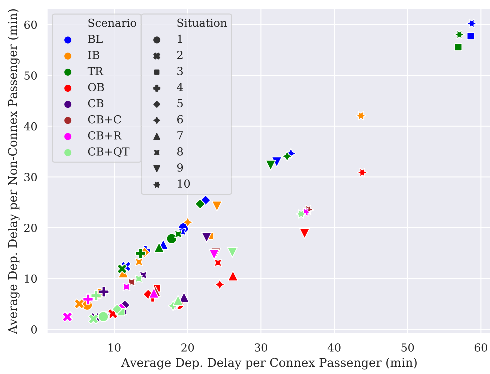

(a) Comparison of total cost and departure delay per flight.




(b) Comparison of total cost and departure delay per passenger.




(c) Comparison of departure delay per flight and passenger.




(d) Comparison of delay for connex and non-connex passengers.

**Fig. 11.** Comparison of key performance indicators against each other.

**Table 6**
Description of available schedule recovery resources per instance of the sensitivity analysis with modified delay cost valuations.

| ID | Scenario Description | Inbound Recovery Resources | | | Transit | Outbound | | |
|---|---|---|---|---|---|---|---|---|
| | | AP | SR/ QB | QP | QT | CP | SC | DD |
| CB | combined inbound, transit and outbound recovery | 1 prio. | 10 CS/ 3 RS | 1 bus | 1 unit | avail. | 1 crew | avail. ( × 1) |
| CB + HC | combined recovery with low delay cost | 1 prio. | 10 CS/ 3 RS | 1 bus | 1 unit | avail. | 1 crew | avail. (× **2**) |
| CB + LC | combined recovery with high delay cost | 1 prio. | 10 CS/ 3 RS | 1 bus | 1 unit | avail. | 1 crew | avail. (× **1/2**) |

*CB* decreases on average to 8.4 min (even below flight delay), while it increases for connecting passengers to 15.8 min (cf. Fig. 11d). Interestingly, separate inbound recovery (*IB*) outperforms outbound recovery (*OB*), given that in *IB* all connections are maintained while stand allocation and transfer times are optimized, whereas in *OB* an increased number of passengers is rebooked onto later flights (cf. Fig. 10). Due to limited flexibility, *OB* generally shows the lowest transfer ratio (i.e., maintained passenger connections) with 91.5% in comparison to 94.6% in *CB* and 95.3% in both instances *CB* +*C* and *CB* + *R*. Thus, because of additional inbound recovery resources in instances *CB* + *C* and *CB* + *R*, 13 passengers more are able to reach their connecting flight, which corresponds to on average 0.9 and 1.1 min delay reductions per passenger respectively (cf. Fig. 9). In contrast, a second quick-turnaround unit in *CB* + *QT* can reduce departure delay per passenger by 0.5 min with the negative side effect of diminishing also the transfer ratio to 94.2% because of more cancelled passenger transfers (cf. Fig. 10).

### *4.3. Sensitivity towards Delay Cost Valuation*

Given the high correlation between average flight delay and total cost, it is important to analyse the sensitivity of the results under the influence of changing delay cost valuations.

Different airlines may have different network structures and recovery policies, which may result in lower or higher marginal cost of departure delay. Thus, two additional scenarios are introduced in Table 6 which resemble resource availability as in the combined recovery scenario *CB*, but with cost of delay multiplied by factor 2 for high cost (*CB* + *HC*) and 0.5 for low cost (*CB* + *LC*).

Due to the variation of delay cost values, it is not expedient to perform a detailed cost comparison with prior instances. In any case, it can be confirmed that delay cost are a major driver of total cost, given that objective values in *CB* + *HC* increase on average by 76% in comparison to *CB* when, *ceteris paribus*, cost of departure delay is doubled. Similarly, when cost of delay is halved in *CB* + *LC*, objective values decrease by 43%.

When the valuation of delay cost is doubled in *CB* + *HC*, Fig. 12 shows a reduction of average flight delay by 0.5 min and almost no notably change in average delay per passenger. Thus, although flights are delayed slightly less due to higher cost of delay (OTP increases from 77% to 79%), there is only limited impact on the overall passenger delay. If connecting and non-connecting passengers are analysed separately, the impact becomes more visible, such that delay reduces for non-connecting passengers by 1.2 min and increases for connecting passengers equally by 1.2 min. The transfer ratio decreases by 0.9% to 93.7%.

The reduced valuation of delay cost causes longer waiting times for transfer passengers and, hence, increases average flight delay by 1.5 min. Thus, OTP decreases by 3% to 74%, whereas the transfer ratio increases by 1.2% to 95.8%. As a result, average departure delay per passenger decreases by 0.6 min. Again, the impact is more visible in the separate analysis, so that delay increases for non-connecting passenger by 0.8 min, while it decreases for connecting passengers by 1.8 min.

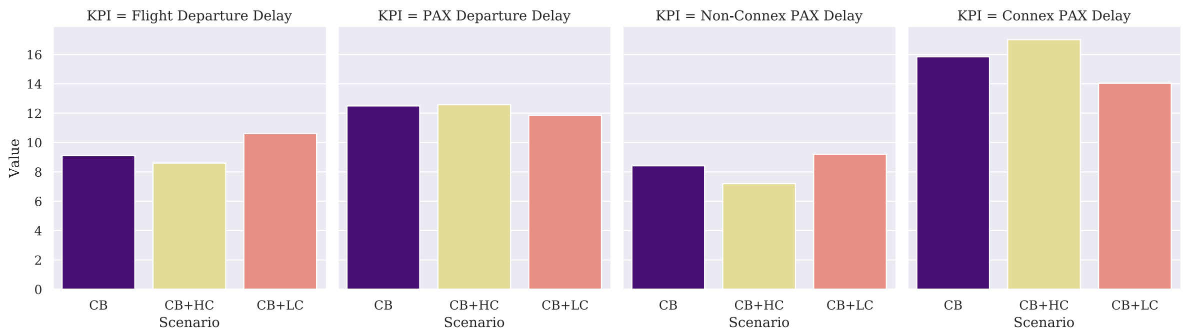



**Fig. 12.** Sensitivity of KPIs average flight and passenger delay with high and low delay cost valuations.

## 5. Conclusion and Future Work

This paper has studied the resilience potential comprised in aircraft ground operations at airline hub airports. For this, a wide range of standardized schedule recovery options is introduced and integrated into a mathematical optimization model in order to calculate an optimal recovery strategy for multiple parallel aircraft turnarounds while considering constrained airport standard and reserve resources.

The application of the model within a case study reveals huge potential for resilience during aircraft ground operations. Thereby, results show that an objective function which aims at minimizing tactical cost concurrently reduces average delay per flight and passenger. Within ten studied arrival delay situations, on average 77% of additional tactical cost can be saved with a combined recovery strategy in comparison to a baseline recovery policy which delays departure flights to maintain all booked passenger transfer connections. At the same time, average flight delay is reduced by roughly 65%, while passengers experience about 53% less delay (despite some passengers being re-booked onto later flights). Note that the unbalance between flight and passenger delay reduction originates from longer waiting times for re-booked transfer passengers, while, on average, non-connecting passengers actually obtain less delay than flights in the combined recovery solution. This confirms findings from previous studies across multiple airlines at an airport (Montlaur and Delgado, 2017).

The developed approach illustrates that the recently studied recovery option to "wait-for-passengers" (Delgado et al., 2016; Marla et al., 2017; Mazzarisi et al., 2019) is by far not the only tactical instrument that airlines have at their disposal during ground operations to minimize cost in case of arrival delays. In fact, the impact of delaying a departure to maintain a connection is very volatile and only efficient in certain network constellations. Instead, results highlight that inbound recovery options, which aim for a redistribution of the available airport infrastructure and reserve resources, provide consistent cost and delay reductions for all studied situations, even if all transfer connections were maintained. Thereby, indirect costs related to the redistribution of resources (e.g., additional workload for supervisors) are currently neglected, such that a freeze horizon should be considered to avoid additional disruptions when the model is implemented in a dynamic setting (e.g., with rolling horizon mode as in Evler et al. (2021)).

The superior performance of a combined recovery strategy in comparison to department-specific solutions highlights the benefits of collaborative decision-making, whereby this paper is one of the first to quantify the inherent resilience potential of integrated schedule recovery during ground operations. It confirms the need that all collaborating stakeholders at an airport share their internal resource schedules, constraints and priorities to standardise (if not automate) the solution generation process across all airline departments to efficiently cope with tactical schedule deviations. The presented approach provides the mathematical and conceptual foundation for a centralized decision support system to be used at an airline's hub control center. Thereby, the model can be adapted to various airport settings and airline schedule recovery policies. It has demonstrated little sensitivity to internal cost valuations and could technically operate also in the absence of accurate cost-definitions, on the basis of commonly-used service quality indicators, such as on-time performance and transfer ratio. However, cost are found to be a viable common denominator to combine ambivalent airline optimization objectives. Thus, further studies are currently in progress which aim at incorporating flight-specific downstream network-dependencies and uncertainties into stochastic delay cost functions (Evler et al., 2020), given that statistically fitted delay cost reference values (Cook, 2015) are not foreseen to prioritise between particular flights.

A parallel study (Evler et al., 2021) has found high stability of the generated recovery solutions towards uncertain arrival times. Such analyses will be continued for further stochastic parameters, such that we aim for a stochastic optimization model, for instance, by including chance constraints as in Asadi et al. (2020). Considering that run time already exceeded the period of one hour while the look-ahead time (LAT) until the first scheduled arrival is set at two hours, we propose a meta-heuristic solution method to speed up the solution finding process and generate solutions for large-scale problems which consist of 30 or more parallel turnarounds (as done in Asadi et al. (2021)).

The analysis of combined recovery scenarios with different additional resources has revealed that, regardless of the additional resource type, similar cost benefits can be achieved because of higher flexibility. These findings might be transferred into more strategic planning stages which evaluate the robustness of the planned schedule with different recovery and resource strategies (cf. the concept of "recoverable robustness" applied by Dijk et al. (2019)). This would enable airlines to limit schedule buffers and negotiate optimal resource access and fees as part of their service level agreements with ground handling service providers or airports. Here, future research should consider how far the model, being strictly airline-focused, is able to comply with the individual objectives of competing airlines and other stakeholders at an airport. Especially for shared resources such as remote stands, catering and fuelling trucks or even runways and ATFM slots, stakeholder-specific slot constraints should be introduced, in which the respective resources are blocked for usage by other airlines. Depending on the airport, this will create more bottleneck situations which may also reduce the flexibility and impact of schedule recovery. It also widens the scope for future research on resource biding or tactical (ATFM) slot swapping, as focused within recently developed SESAR mechanisms, such as UDPP (Pilon et al., 2016). Finally, the model should be validated in a real operational environment to study the influence of daily changing aircraft, crew and passenger itineraries on the recovery performance.

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